Conformal Quantile Nowcasting

Martin Fankhauser (Bocconi University, PhD candidate)

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Research Question

Quantile nowcasts of GDP growth summarize macroeconomic tail risk in real time (Adrian, Boyarchenko, and Giannone (2019), Carriero, Clark, and Marcellino (2022)), but producing them is costly:

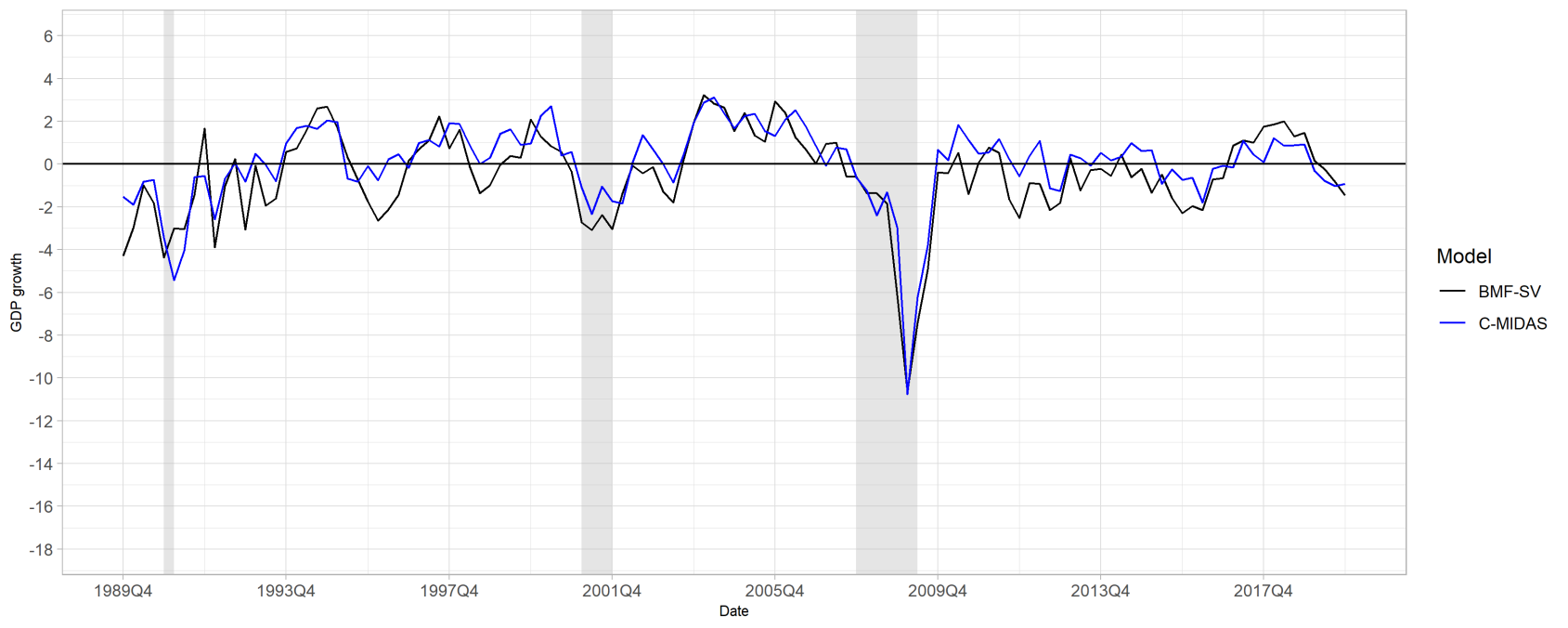
- **Stochastic volatility** (BMF-SV) Carriero, Clark, and Marcellino (2015)
  - many latent states ⇒ slow
- **Residual bootstrap** for MIDAS Aastveit, Foroni, and Ravazzolo (2017)
  - re-estimation in every replication, fails with many regressors

Can we turn point nowcasts into calibrated quantile nowcasts at (almost) no cost?

⇒ Yes, using conformal prediction:

α-quantile nowcast = point nowcast + empirical α-quantile of past signed errors

Preview Results: Fast Nowcasting



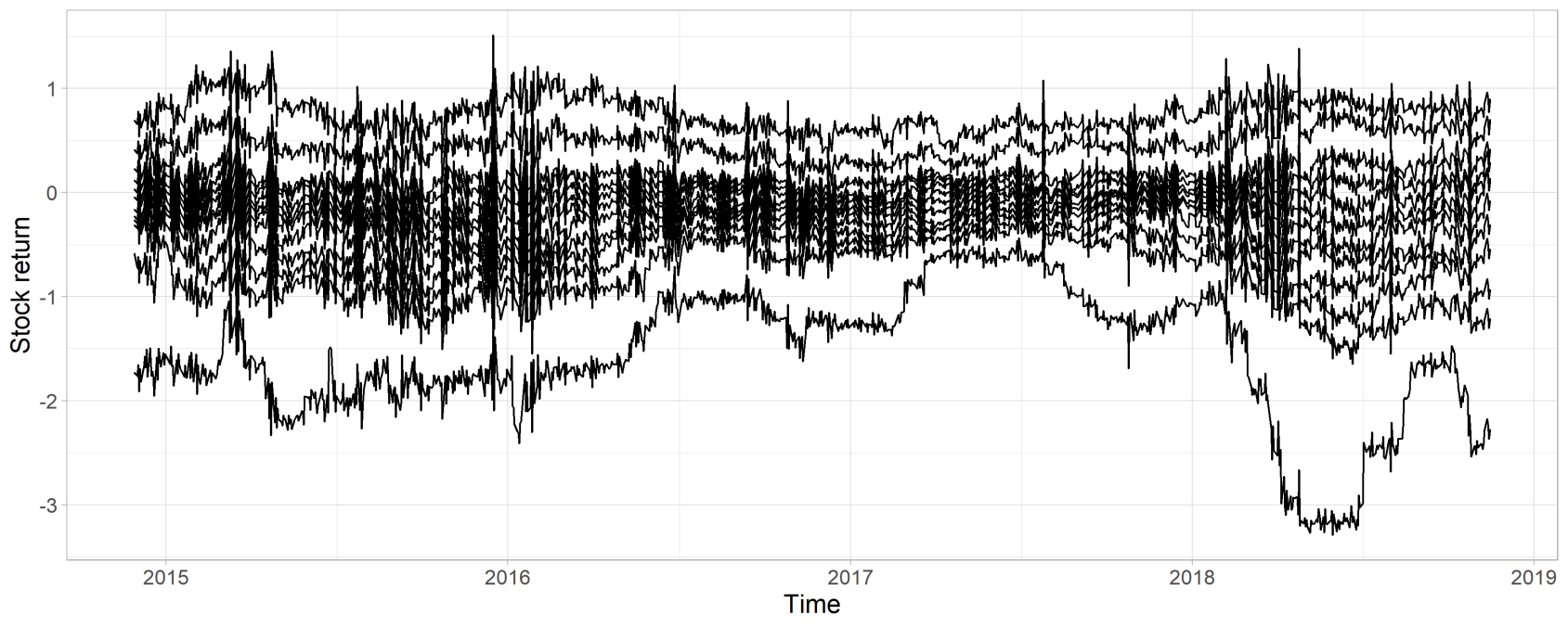
Real-time 5% quantile nowcasts of US GDP growth, 1989Q4–2019Q3, week 9.

C-MIDAS: conformalized MIDAS (CP1-MIDAS).

Coverage and quantile scores comparable to the BMF-SV (Carriero, Clark, and Marcellino (2015)), at essentially the cost of the point nowcast.

Covid: Robust to Outliers

Preview Results: No Quantile Crossing



One-step-ahead 5%, 10%, . . . , 95% quantile forecasts of daily Microsoft returns from a conformalized AR(4).

Quantile losses within 3% of the dynamic multiple quantile (DMQ) model of Catania and Luati (2023), **non-crossing by construction**.

Full Results

Literature and Contribution

Conformal Prediction

Vovk, Gammerman, and Shafer (2005), Linusson, Johansson, and Löfström (2014), Romano, Patterson, and Candès (2019), Chernozhukov, Wüthrich, and Zhu (2021)

⇒ **One-sided conformal quantiles from point forecasts alone**

Conformal Prediction for Dependent Data

Gibbs and Candès (2024), Barber et al. (2023), Oliveira et al. (2024)

⇒ **Approximate validity, benchmarks**

Intervals from Past Forecast Errors

Williams and Goodman (1971), Lee and Scholtes (2014), Knüppel (2014)

⇒ **Conditions for calibration and a precise target**

Quantile Nowcasting and Bootstrap

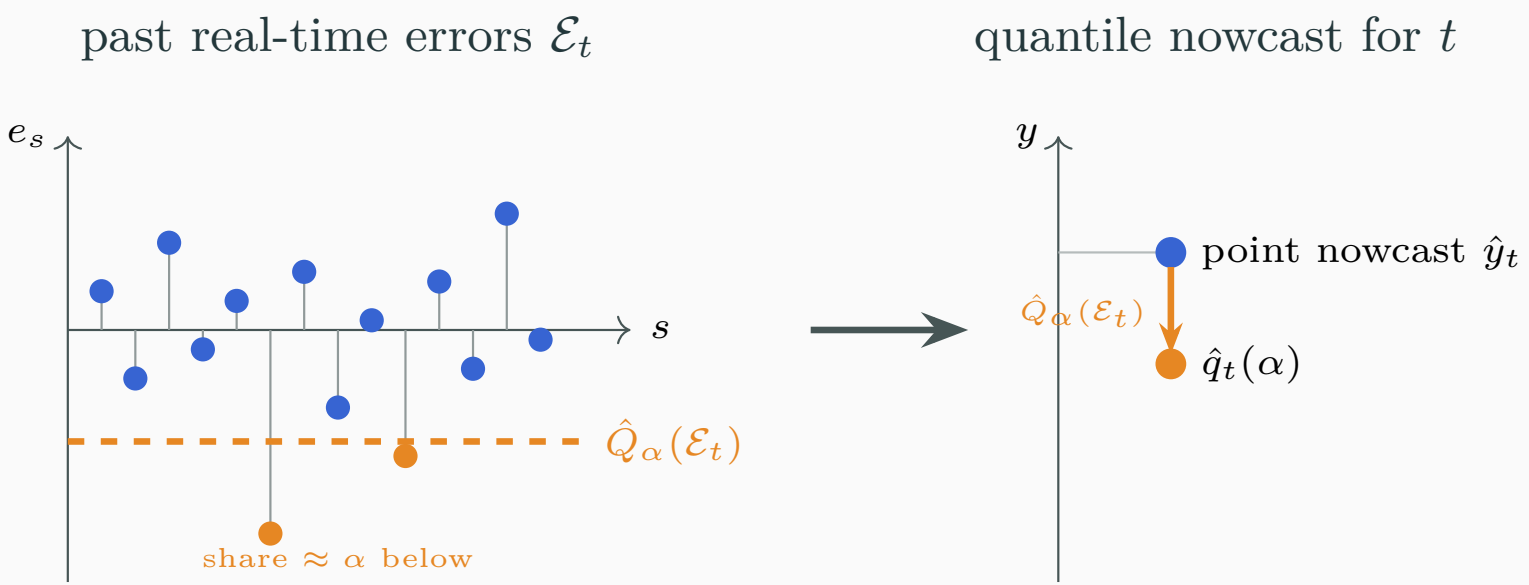
Aastveit, Foroni, and Ravazzolo (2017), Carriero, Clark, and Marcellino (2022), Pan and Politis (2016), Catania and Luati (2023)

⇒ **Fast, model-agnostic alternative**

Conformal Quantile Nowcasts

The Idea

Any point nowcast $\hat{y}_t$ (MIDAS, BMF, ML, ...). Once y_t is published, record the signed error $e_t = y_t - \hat{y}_t$.



Definition 1: Conformal quantile nowcast (CP1)

$$\hat{q}_t(\alpha) = \hat{y}_t + \hat{Q}_\alpha(\mathcal{E}_t)$$

Algorithm

Algorithm 1 Conformal quantile nowcasting

Input: levels $\alpha_1 < \cdots < \alpha_K$; method $\mathcal{A}$; publication lag ℓ; window w

for $t = s_0, \ldots, T$ **do**
  Point nowcast $\hat{y}_t$ from $\mathcal{A}$ estimated on $\mathcal{I}_t$
  if $t \geq t_0$ **then**
    $\mathcal{E}_t \leftarrow \{e_s : t - \ell - w < s \leq t - \ell,\ s \geq s_0\}$ ▷ published errors
    $\hat{q}_t(\alpha_k) \leftarrow \hat{y}_t + \hat{Q}_{\alpha_k}(\mathcal{E}_t),\ \ k = 1, \ldots, K$
  end if
  Once y_t is published: $e_t \leftarrow y_t - \hat{y}_t$ ▷ signed error
end for

Output: $\hat{q}_t(\alpha_1) \leq \cdots \leq \hat{q}_t(\alpha_K)$

Cost = point nowcast + sorting n_t numbers

no extra model, no re-estimation on resampled data

Finite-Sample Calibration

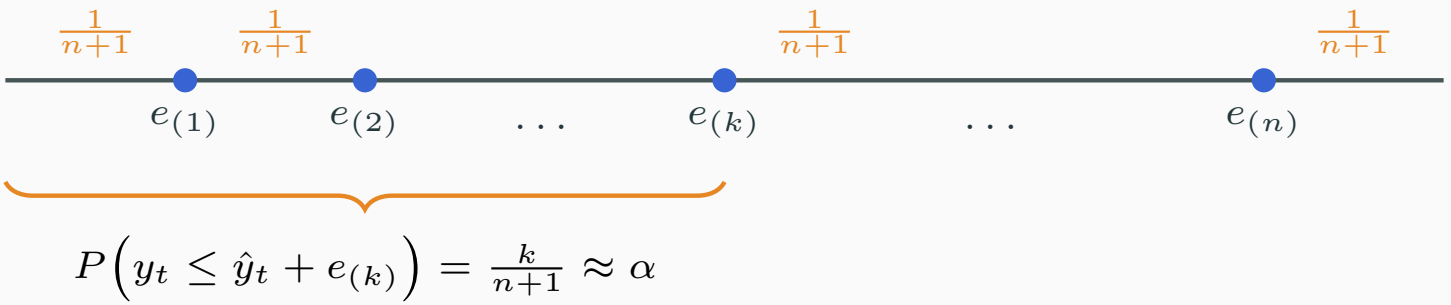
Proposition 1

If the $n+1$ errors in $\mathcal{E}_t \cup \{e_t\}$ are exchangeable:

(i) **calibration:** $\left| P\big(y_t \leq \hat{q}_t(\alpha)\big) - \alpha \right| \leq \frac{2}{n+1}$

(ii) **no crossing:** $\alpha < \alpha' \Rightarrow \hat{q}_t(\alpha) \leq \hat{q}_t(\alpha')$

Why? The rank of e_t among the $n+1$ errors is uniform:

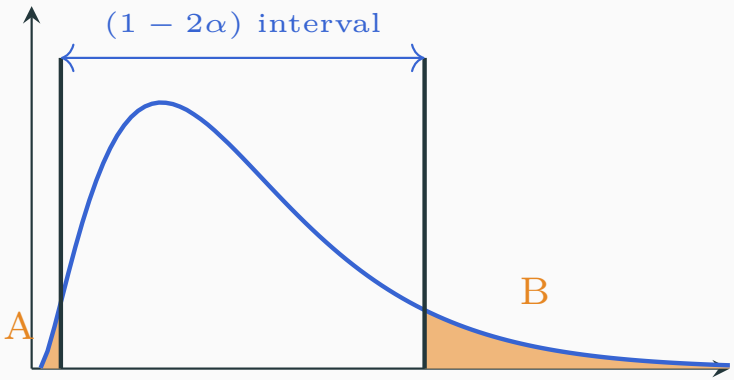


$\Rightarrow$ no assumption on the error distribution or the nowcasting method

Full Statement and Proof

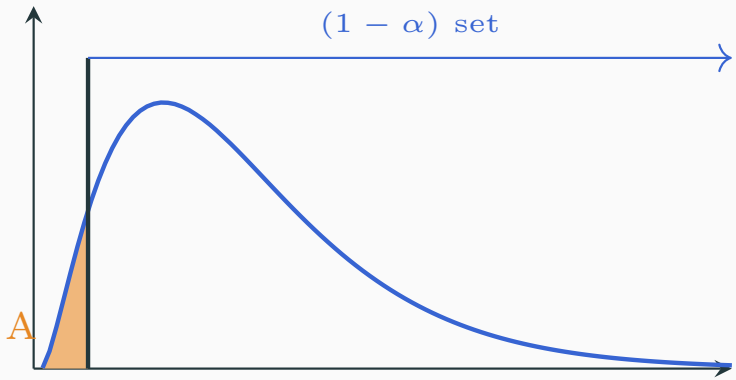
Why Signed Errors?

absolute errors $|e_s|$ (CP2)



$P(A) \approx 0.01, \quad P(B) \approx 0.09$

signed errors e_s (CP1)



$P(A) = 0.05$

Skewed errors (shifted $\chi^2(5)$), $\alpha = 0.05$

- Two-sided intervals control only the total miscoverage 2α
- Signed errors control each tail (Linusson, Johansson, and Löfström (2014))

Benchmark CP2: lower end of the two-sided interval.

Beyond Exchangeability

Errors	Guarantee for CP1
exchangeable	exact in finite samples (Proposition)
β-mixing (e.g. ARMA, GARCH)	approximate, penalty vanishes as n grows Oliveira et al. (2024)
arbitrary shifts	adaptive levels (FACI), but quantiles can cross Gibbs and Candès (2024)

Target: the marginal quantile (conditional if $e_t \perp\!\!\!\perp$ predictors)
$\Rightarrow$ How much is lost relative to SV models is an empirical question

Exchangeability and Number of Errors | FACI

Monte Carlo Simulations

AR(1) with Normal Errors

	$\alpha = 0.01$			$\alpha = 0.05$			$\alpha = 0.10$		
	$T = 50$	$T = 100$	$T = 200$	$T = 50$	$T = 100$	$T = 200$	$T = 50$	$T = 100$	$T = 200$
CP1	0.036	0.019	0.014	0.076	0.060	0.054	0.124	0.110	0.103
CP2	0.021	0.015	0.012	0.063	0.053	0.051	0.109	0.103	0.101
FpB	0.017	0.014	0.012	0.057	0.053	0.052	0.104	0.105	0.102
FsB	0.015	0.013	0.012	0.054	0.052	0.051	0.102	0.103	0.102
PI-CP	0.957	0.971	0.977	0.877	0.895	0.898	0.784	0.797	0.800
PI-FpB	0.965	0.972	0.975	0.889	0.895	0.895	0.793	0.794	0.797
PI-FsB	0.970	0.974	0.976	0.894	0.897	0.897	0.797	0.797	0.798

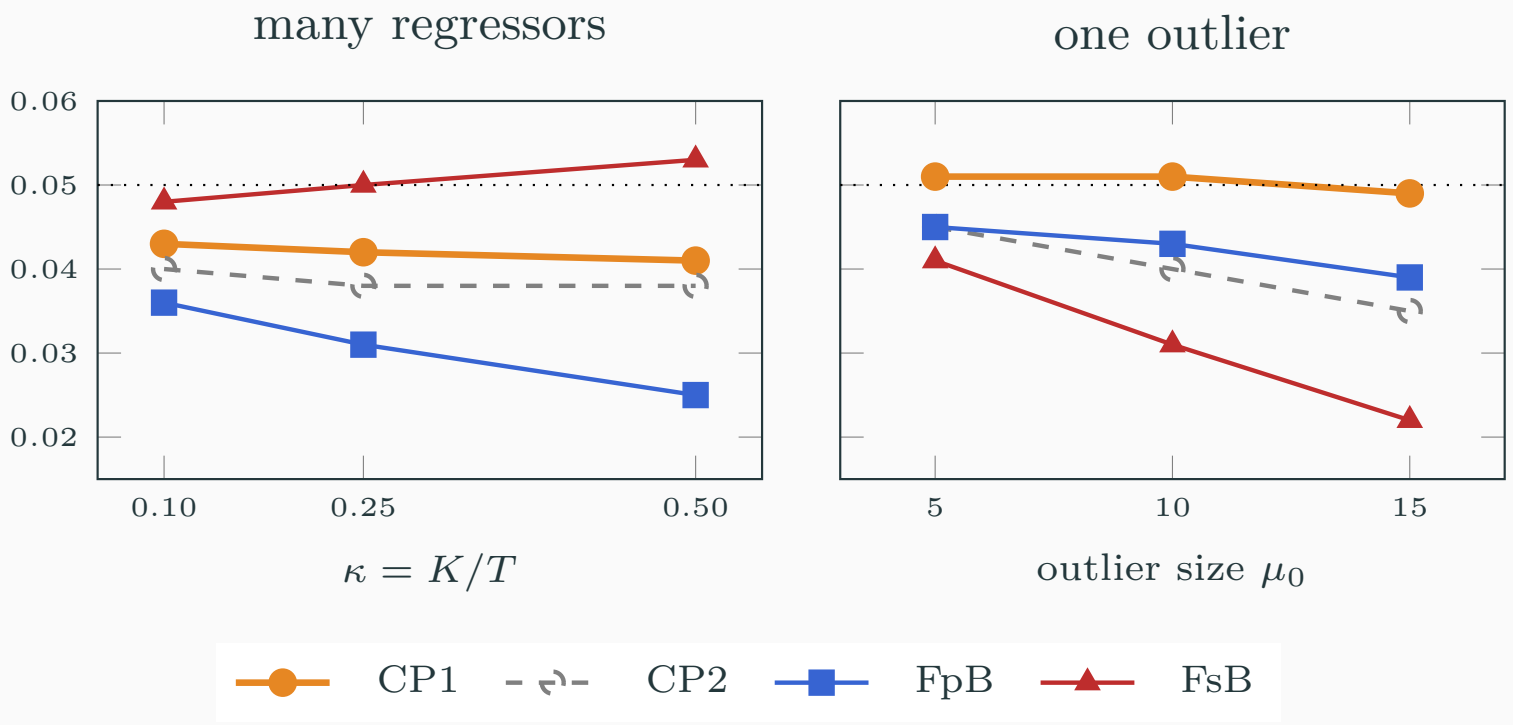
$y_t = 0.5y_{t-1} + \epsilon_t$, $\epsilon_t \sim N(0, 1)$. Exceedance frequencies; PI rows: coverage of two-sided $(1 - 2\alpha)$ intervals. FpB/FsB: forward bootstraps of Pan and Politis (2016).

- Bootstraps closer to nominal for small T and extreme α
- CP1 as predicted: $T = 50$, $\alpha = 0.01 \Rightarrow [0.032, 0.065]$

Skewed χ^2 Errors

Many Regressors and Outliers

Exceedance frequencies at $\alpha = 0.05$ (dotted: nominal level), $T = 100$



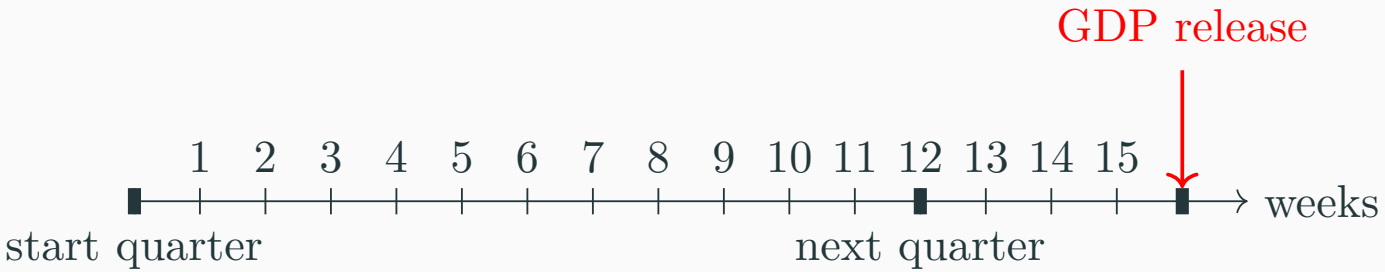
- Many regressors: non-studentized bootstrap (FpB) overcovers
- Outlier: studentized bootstrap (FsB) overcovers, CP1 unaffected

Full Table: Regressors | Full Table: Outliers

Nowcasting US GDP Tail Risk

Weekly Nowcasting

15 nowcasts per quarter, before the first GDP release:



- Data and design of Carriero, Clark, and Marcellino (2022)
- CP1-MIDAS vs. BMF-SV (also CP2, FACI, DCP-BMF, BOOT-MIDAS)
- Real time, out of sample: 1989Q4–2019Q3 (120 quarters)

Results: Coverage

	Weeks							
	1	3	5	7	9	11	13	15
BMF-SV	0.066	0.066	0.05	0.033	0.05	0.05	0.033	0.033
CP1-MIDAS	0.033	0.033	0.041	0.058	0.05	0.05	0.074	0.05
CP2-MIDAS	0.033	0.041	0.033	0.017 ***	0.017 ***	0.033	0.05	0.041
BOOT-MIDAS	0.017 ***	0.008 ***	0.017 ***	0.017 ***	0.017 ***	0.017 ***	0.008 ***	0.008 ***
DCP-BMF	0.099 *	0.099 *	0.124 **	0.091	0.091	0.099 *	0.099 *	0.099 *
FACI-MIDAS	0.025 *	0.025 *	0.033	0.041	0.041	0.025 *	0.041	0.025 *

Exceedance frequencies of 5% quantile nowcasts, odd weeks. *, **, * * *: rejection of correct coverage at 10%, 5%, 1%.

- BMF-SV and CP1-MIDAS: never rejected
- CP2-MIDAS: too few exceedances in some weeks
- BOOT-MIDAS: far too few exceedances in every week

Even Weeks Other Bootstraps

Results: Quantile Scores

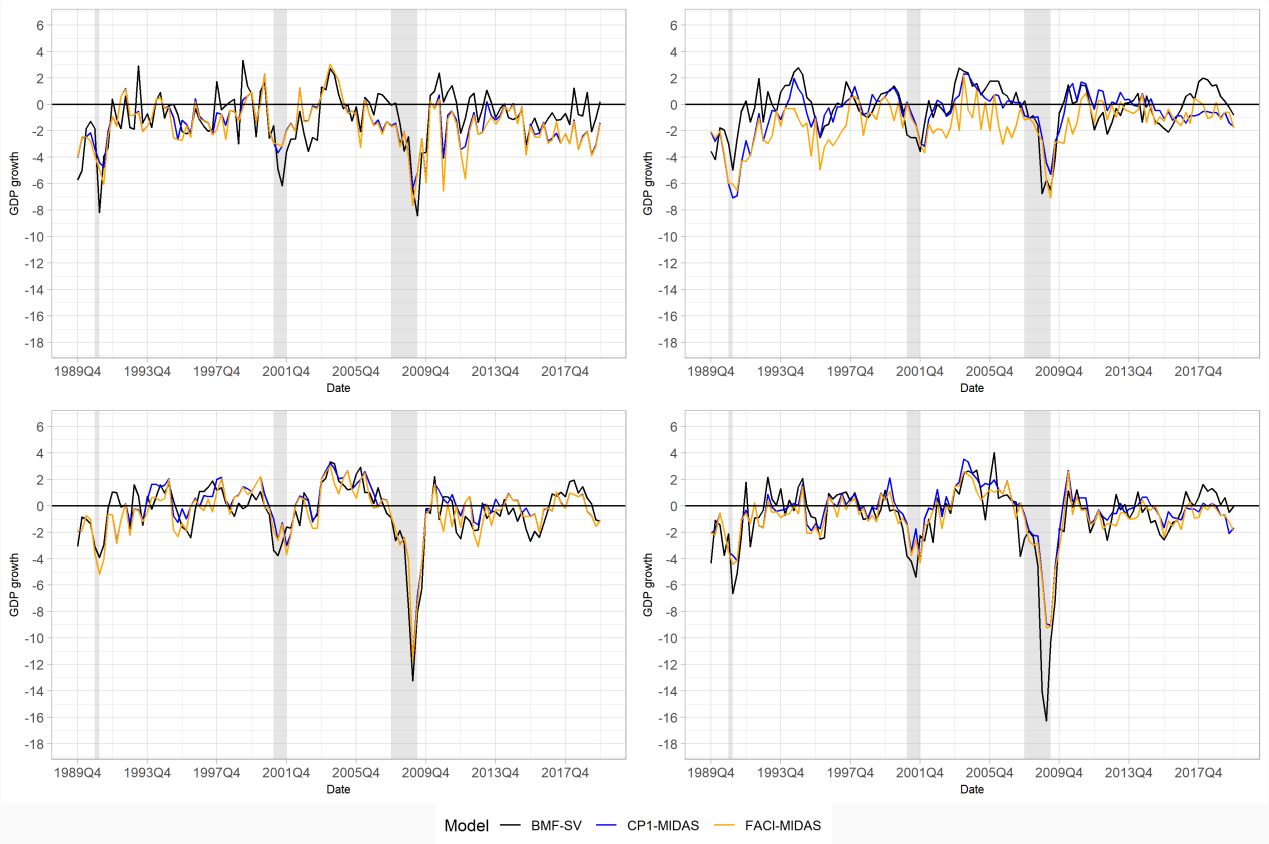
	Weeks							
	1	3	5	7	9	11	13	15
BMF-SV	0.252	0.244	0.176	0.162	0.19	0.174	0.192	0.205
CP1-MIDAS	0.908	0.853	1.331	1.136	0.916	1.053	0.846	0.901
CP2-MIDAS	0.893	0.842	1.324 **	1.325 ***	0.966	1.043	0.826	0.892
BOOT-MIDAS	1.080	0.982	1.452 ***	1.494 ***	1.285 ***	1.362 ***	1.325 ***	1.162 **
DCP-BMF	1.209	1.242	1.42	1.327	1.114	1.135	1.047	0.948
FACI-MIDAS	0.938	0.934	1.495 **	1.216 *	0.942	1.085	0.806	0.922

First row: average quantile score of the BMF-SV; other rows relative to it (shaded: lower loss). Stars: Diebold–Mariano test.

- CP1-MIDAS lower in 9 of 15 weeks, never significantly different
- BOOT-MIDAS significantly worse in most weeks

Even Weeks

Quantile Nowcasts within the Quarter



Weeks 1, 5, 10 and 15 (left to right, top to bottom). Shaded: NBER recessions.

Later in the quarter the point nowcast drives the time variation.

Covid

Conclusion

In this paper we ...

... propose **conformal quantile nowcasting**

- point nowcast + empirical quantile of past signed errors
- calibrated in finite samples, no quantile crossing

... compare it with the **bootstrap** in simulations

- robust to many regressors and outliers

... show that it is competitive in **applications**

- US GDP tail risk: comparable to the BMF-SV
- Microsoft percentiles: on par with the DMQ

Complements rather than replaces models of the conditional distribution.

Proposition: Full Statement

Proposition 1: Finite-sample calibration

Fix t and suppose the $n+1$ errors in $\mathcal{E}_t \cup \{e_t\}$ are exchangeable and a.s. distinct, with order statistics $e_{(1)} < \cdots < e_{(n)}$ of $\mathcal{E}_t$.

(i) $P\big(y_t \le \hat{y}_t + e_{(k)}\big) = \frac{k}{n+1}$ for every $k \in \{1, \ldots, n\}$

(ii) conservative $\hat{Q}_\alpha = e_{(\lfloor \alpha(n+1) \rfloor)}$, $\alpha \ge \frac{1}{n+1}$:
$\alpha - \frac{1}{n+1} < P\big(y_t \le \hat{q}_t(\alpha)\big) \le \alpha$

(iii) interpolated $\hat{Q}_\alpha$ (type 7 in R), $j = \lfloor (n-1)\alpha \rfloor + 1$:
$\frac{j}{n+1} \le P\big(y_t \le \hat{q}_t(\alpha)\big) \le \frac{j+1}{n+1}$

(iv) $\alpha < \alpha' \Rightarrow \hat{q}_t(\alpha) \le \hat{q}_t(\alpha')$ for every realization of the data

(iii) implies $|P(y_t \le \hat{q}_t(\alpha)) - \alpha| \le \frac{2}{n+1}$. All computations use the interpolated version.

Proof Back

Proof

Let R be the rank of e_t among the $n+1$ errors in $\mathcal{E}_t \cup \{e_t\}$. Exchangeable and distinct errors $\Rightarrow$ R is uniform on $\{1, \ldots, n+1\}$.

(i) Since $y_t - \hat{y}_t = e_t$:

$$y_t \le \hat{y}_t + e_{(k)} \iff e_t \le e_{(k)} \iff R \le k \quad \Rightarrow \quad P\big(y_t \le \hat{y}_t + e_{(k)}\big) = \tfrac{k}{n+1}$$

(ii) Conservative: $k = \lfloor \alpha(n+1) \rfloor$ satisfies $\alpha(n+1) - 1 < k \le \alpha(n+1)$, apply (i).

(iii) Interpolated: $\hat{Q}_\alpha(\mathcal{E}_t) \in [e_{(j)}, e_{(j+1)}]$, so

$$\{y_t \le \hat{y}_t + e_{(j)}\} \subseteq \{y_t \le \hat{q}_t(\alpha)\} \subseteq \{y_t \le \hat{y}_t + e_{(j+1)}\}$$

and (i) gives the bounds. Since $(n-1)\alpha < j \le (n-1)\alpha + 1$, both are within $\frac{2}{n+1}$ of α.

(iv) $\hat{Q}_\alpha(\mathcal{E})$ is non-decreasing in α, and all levels share $\hat{y}_t$ and $\mathcal{E}_t$. □

Back

Exchangeability and Number of Errors

When are the errors exchangeable?

- iid data and $\mathcal{A}$ estimated once on a training sample (split conformal)
- In nowcasting: serial dependence and re-estimation every period $\Rightarrow$ only approximately, assessed in simulations and applications

How many past errors?

- Calibrated α-quantile requires $n \geq 1/\alpha - 1$
- Distinct forecasts of all percentiles require at least 100 past errors

Back

Beyond Exchangeability: (F)ACI

Adaptive conformal inference (Gibbs and Candes (2021)) treats the level as a parameter updated online:

$$\text{err}_t = \mathbf{1}\{y_t \leq \hat{q}_t(\alpha_t)\}, \qquad \alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$$

$\Rightarrow$ level moves outward after an exceedance and inward otherwise.

Fully adaptive conformal inference (FACI) (Gibbs and Candès (2024))

- aggregates over a grid of step sizes γ
- coverage over arbitrary time intervals, no assumptions on the DGP
- levels updated separately per quantile $\Rightarrow$ no monotonicity guarantee

In the applications: FACI wrapped around CP1. Back

AR(1) with Skewed Errors

	$\alpha = 0.01$			$\alpha = 0.05$			$\alpha = 0.10$		
	$T = 50$	$T = 100$	$T = 200$	$T = 50$	$T = 100$	$T = 200$	$T = 50$	$T = 100$	$T = 200$
CP1	0.033	0.016	0.012	0.069	0.054	0.050	0.119	0.104	0.099
CP2	0.001	0	0	0.022	0.012	0.009	0.080	0.073	0.072
FpB	0.007	0.007	0.009	0.042	0.047	0.049	0.096	0.099	0.100
FsB	0.005	0.006	0.008	0.041	0.046	0.048	0.095	0.098	0.099
PI-CP	0.957	0.971	0.976	0.884	0.899	0.903	0.788	0.802	0.804
PI-FpB	0.974	0.978	0.978	0.900	0.900	0.899	0.797	0.799	0.799
PI-FsB	0.978	0.980	0.980	0.906	0.903	0.902	0.802	0.802	0.801

$y_t = 0.5y_{t-1} + \epsilon_t$, ϵ_t from a $\chi^2(5)$ distribution shifted to mean zero. Models as in the main text.

- CP1 performs as with normal errors
- CP2 heavily miscalibrated: about 0 for $\alpha = 0.01$, 0.01–0.02 for $\alpha = 0.05$

Why Signed Errors? Back

Many Exogenous Regressors: Full Table

	$\alpha = 0.01$			$\alpha = 0.05$			$\alpha = 0.10$		
	$\kappa = 0.10$	$\kappa = 0.25$	$\kappa = 0.5$	$\kappa = 0.10$	$\kappa = 0.25$	$\kappa = 0.5$	$\kappa = 0.10$	$\kappa = 0.25$	$\kappa = 0.5$
CP1	0.013	0.011	0.008	0.043	0.042	0.041	0.105	0.084	0.073
CP2	0.010	0.008	0.006	0.040	0.038	0.038	0.099	0.076	0.066
FpB	0.008	0.007	0.001	0.036	0.031	0.025	0.102	0.076	0.047
FsB	0.010	0.012	0.010	0.048	0.050	0.053	0.112	0.100	0.084
PI-CP	0.981	0.987	0.988	0.910	0.923	0.928	0.810	0.838	0.859
PI-FpB	0.986	0.987	0.996	0.917	0.937	0.953	0.808	0.839	0.903
PI-FsB	0.983	0.979	0.982	0.896	0.899	0.899	0.793	0.797	0.824

$y_t = 0.5y_{t-1} + X'_{t-1}\beta + \epsilon_t$, $\epsilon_t \sim N(0,1)$, $T = 100$, K regressors, $\kappa = K/T$. Mimics a MIDAS regression with many high-frequency regressors.

- FpB overcovers increasingly in κ: 0.047 for $\alpha = 0.10$ at $\kappa = 0.5$
- FsB repairs the problem, CP1 somewhat conservative as κ grows

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Outliers: Full Table

	Standard Normal Errors $\epsilon_t \sim N(0,1)$ with 1% outliers from $-\lvert N(\mu_0, 0.5)\rvert$								
	$\alpha = 0.01$			$\alpha = 0.05$			$\alpha = 0.10$		
	$\mu_0 = 5$	$\mu_0 = 10$	$\mu_0 = 15$	$\mu_0 = 5$	$\mu_0 = 10$	$\mu_0 = 15$	$\mu_0 = 5$	$\mu_0 = 10$	$\mu_0 = 15$
CP1	0.009	0.009	0.009	0.051	0.051	0.049	0.095	0.097	0.097
CP2	0.009	0.007	0.008	0.045	0.040	0.035	0.086	0.080	0.072
FpB	0.006	0.006	0.006	0.045	0.043	0.039	0.090	0.090	0.087
FsB	0.005	0.005	0.005	0.041	0.031	0.022	0.086	0.078	0.064
PI-CP	0.981	0.983	0.980	0.902	0.905	0.908	0.808	0.808	0.807
PI-FpB	0.979	0.981	0.978	0.902	0.905	0.911	0.805	0.807	0.811
PI-FsB	0.982	0.988	0.987	0.908	0.925	0.941	0.812	0.829	0.847

$y_t = 0.5y_{t-1} + \epsilon_t$, $T = 100$, one error in the sample drawn from $-|N(\mu_0, 0.5)|$, future errors from the corresponding mixture.

- Studentized bootstrap scales residuals by an outlier-driven standard deviation
- CP1: an outlier shifts $\hat{Q}_\alpha(\mathcal{E}_t)$ by at most one order statistic

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Nowcasting: Even Weeks

Exceedance frequencies of 5% quantile nowcasts

	Weeks						
	2	4	6	8	10	12	14
BMF-SV	0.058	0.066	0.041	0.041	0.066	0.033	0.05
CP1-MIDAS	0.025 *	0.033	0.066	0.066	0.041	0.058	0.058
CP2-MIDAS	0.033	0.033	0.025 *	0.017 ***	0.017 ***	0.033	0.05
BOOT-MIDAS	0.017 ***	0.017 ***	0.017 ***	0.017 ***	0.008 ***	0.008 ***	0.008 ***
DCP-BMF	0.091	0.083	0.107 **	0.099 *	0.099 *	0.099 *	0.083
C-MIDAS	0.033	0.033	0.041	0.033	0.025 *	0.033	0.05

Quantile scores (BMF-SV) and ratios relative to the BMF-SV

	Weeks						
	2	4	6	8	10	12	14
BMF-SV	0.254	0.225	0.17	0.174	0.188	0.167	0.204
CP1-MIDAS	0.857	0.881	1.269	1.078	0.896	1.064	0.876
CP2-MIDAS	0.808 **	0.882	1.39 ***	1.222 ***	0.929	1.057	0.872
BOOT-MIDAS	1.014	1.066	1.461 ***	1.384 ***	1.25 ***	1.388 ***	1.164 **
DCP-BMF	1.133	1.155	1.392	1.249	1.086	1.235	0.995
C-MIDAS	0.939	0.915	1.38 *	1.207 *	0.915	1.061	0.89

1989Q4–2019Q3. C-MIDAS: FACI-MIDAS. Stars as in the main tables.

Only rejection for CP1-MIDAS: 10% level in week 2.

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Nowcasting: Alternative Bootstraps

	Weeks							
	1	3	5	7	9	11	13	15
BOOT-MIDAS	0.017 ***	0.008 ***	0.017 ***	0.017 ***	0.017 ***	0.017 ***	0.008 ***	0.008 ***
FpB	0.017 ***	0.008 ***	0.017 ***	0 ***	0 ***	0 ***	0 ***	0.008 ***
FsB	0.017 ***	0.017 ***	0.025 *	0.008 ***	0.017 ***	0.017 ***	0.017 ***	0.008 ***

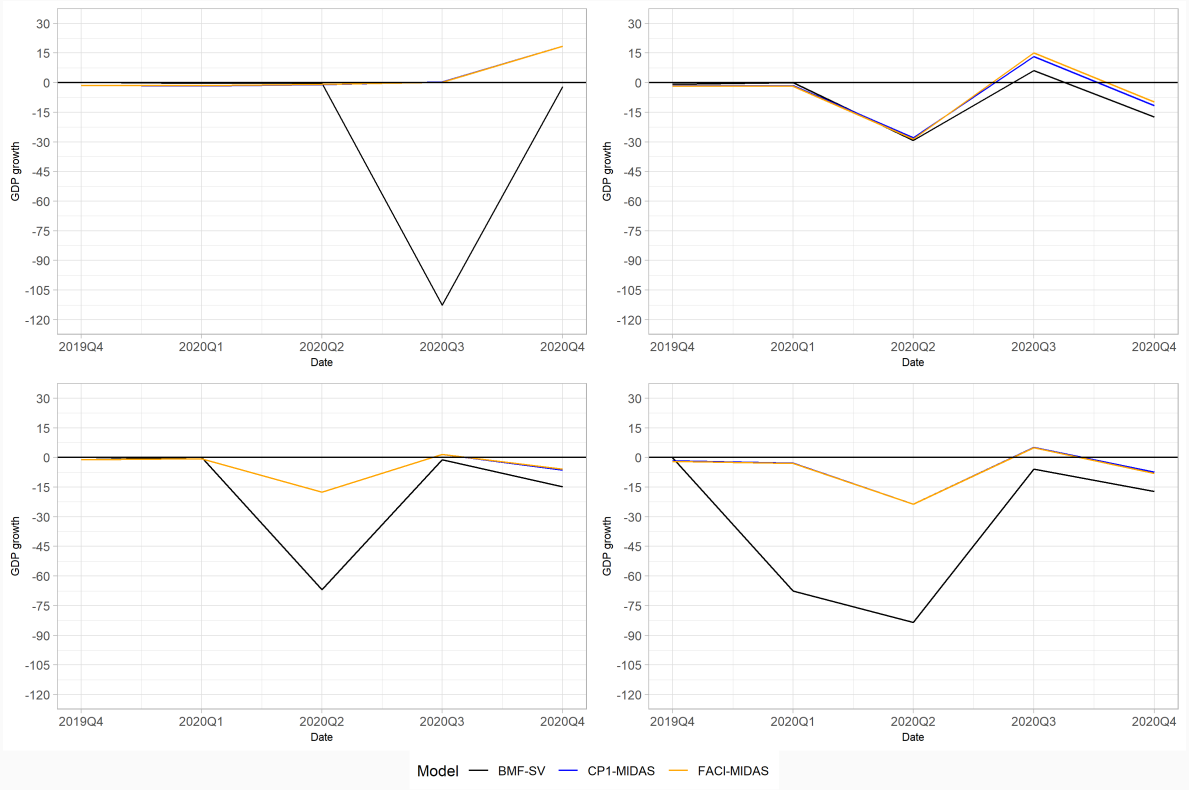
	Weeks						
	2	4	6	8	10	12	14
BOOT-MIDAS	0.017 ***	0.017 ***	0.017 ***	0.017 ***	0.008 ***	0.008 ***	0.008 ***
FpB	0.008 ***	0 ***	0 ***	0 ***	0 ***	0 ***	0 ***
FsB	0.017 ***	0.017 ***	0.025 *	0.008 ***	0.017 ***	0.017 ***	0.017 ***

Exceedance frequencies of 5% quantile nowcasts. BOOT-MIDAS: Aastveit, Foroni, and Ravazzolo (2017); FpB/FsB: forward bootstraps of Pan and Politis (2016) applied to the MIDAS model.

All three bootstraps overcover in every week, in line with the simulations.

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Covid-19 Period



Real-time 5% quantile nowcasts, 2019Q4–2020Q4, weeks 1, 5, 10 and 15 (left to right, top to bottom).

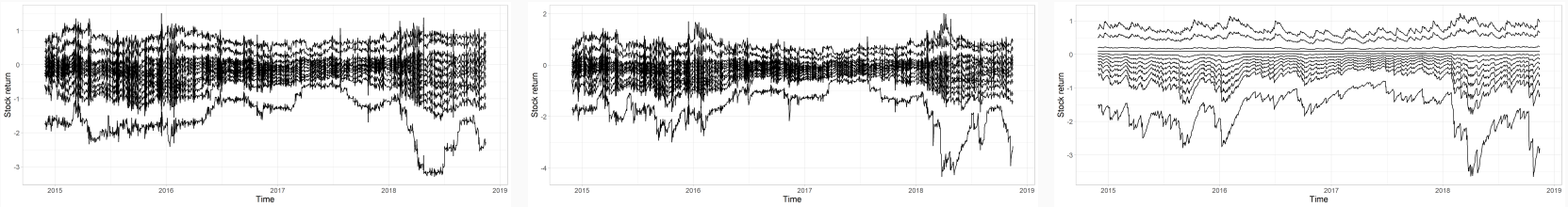
BMF-SV reacts strongly to the extreme observations, conformal quantile nowcasts do not: past errors enter only through their rank.

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Application:
Multiple Quantile Forecasting

Forecasting All Percentiles of Microsoft Returns

- Daily log returns of Microsoft, Dec 2010 – Nov 2018, as in Catania and Luati (2023)
- DMQ estimated once on the first 1,000 observations
- CP1: AR(4) re-estimated every day, rolling window of $w = 100$ errors



One-step-ahead 5%, 10%, . . . , 95% quantile forecasts. Left to right: CP1, FACI, DMQ.

Re-estimating the AR(4) every day takes seconds, re-estimating the DMQ every day would be prohibitively expensive.

Results: Quantile Loss

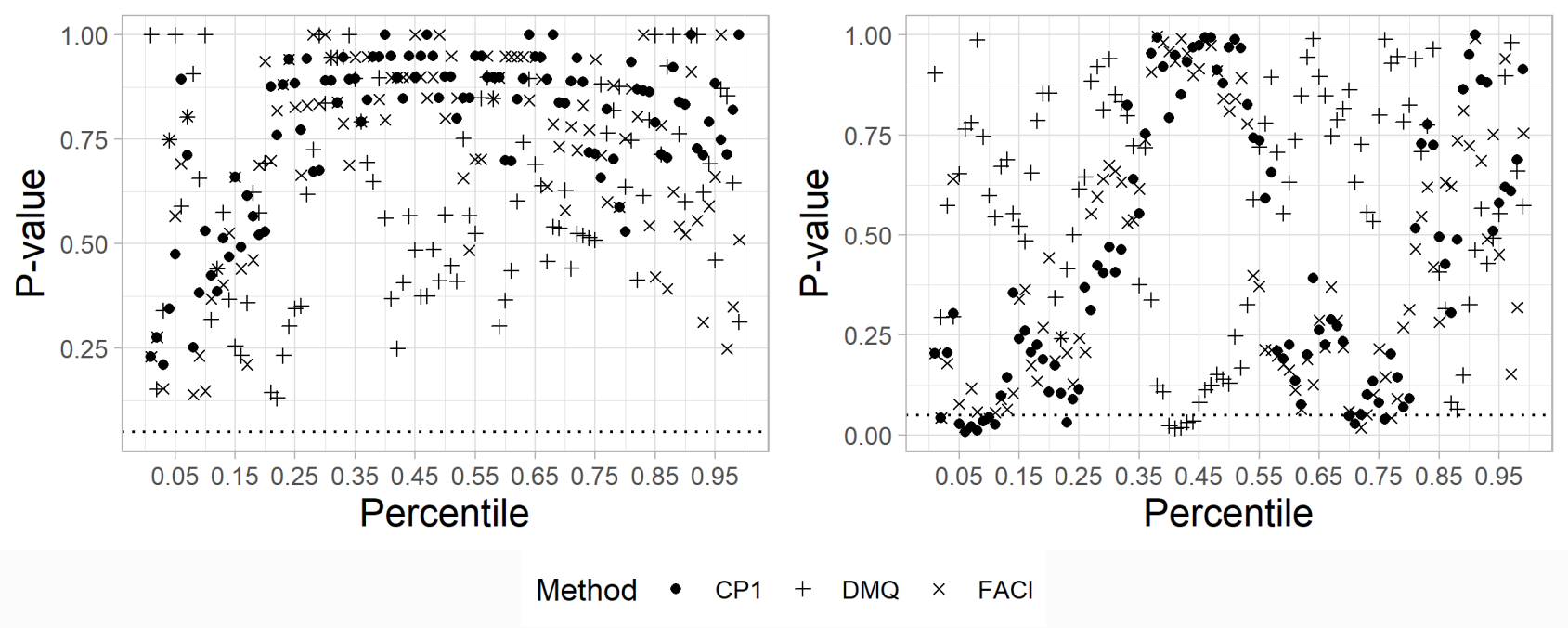
	Aggregate Percentiles									
	[0.01,0.10)	[0.10,0.20)	[0.20,0.30)	[0.30,0.40)	[0.40,0.50)	[0.50,0.60)	[0.60,0.70)	[0.70,0.80)	[0.80,0.90)	[0.90,0.99)
FACI	0.987	0.994	0.995	0.994	0.995	0.992	0.987	0.980	0.971	0.953
CP1	1.003	0.997	0.998	0.994	0.996	0.995	0.992	0.987	0.984	0.973

Sum of the percentile losses within each decile relative to the DMQ, values below one favor the conformal methods.

- CP1 within 3% of the DMQ in every decile, FACI lower in every decile
- The 75% model confidence set (Hansen, Lunde, and Nason (2011)) contains all three methods for the first five deciles and only FACI for the remaining ones

Backtests | Simulation: DMQ as DGP | Window Length

Results: Backtests



p-values of the **unconditional** (left) and **conditional** (right) coverage tests of Kupiec (1995) and Christoffersen (1998) for each percentile.

All methods pass the unconditional test. Conditional test: CP1 rejected only slightly more often than the DMQ, FACI best but no monotonicity guarantee. Back

Simulation: True DGP is DMQ

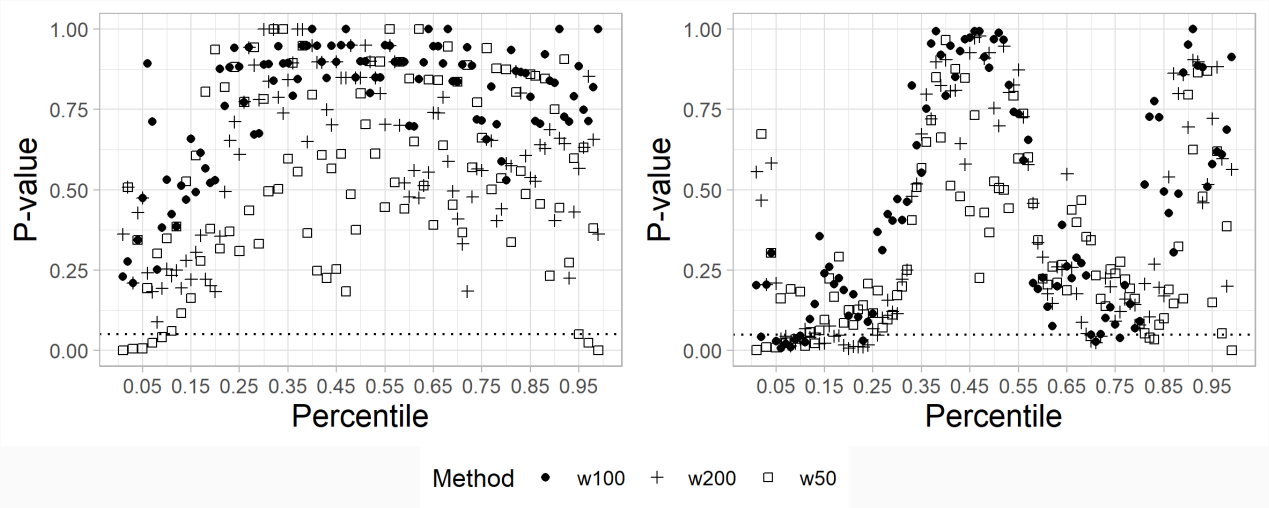
	Deciles								
	α=0.10	α=0.20	α=0.30	α=0.40	α=0.50	α=0.60	α=0.70	α=0.80	α=0.90
n=50	1.055	1.033	1.038	1.050	1.051	1.053	1.054	1.036	1.046
n=100	1.014	1.009	1.015	1.010	1.014	1.027	1.025	1.024	1.024
n=200	1.060	1.030	1.019	1.020	1.030	1.015	0.995	0.999	0.996

Quantile loss of CP1 (AR(4) point forecast) relative to the DMQ for each decile, data generated by the DMQ of Catania and Luati (2023), 500 replications. Values below one favor CP1.

Even if the data are generated by the DMQ and the DMQ is re-estimated in every replication, CP1 loses at most 6%. Back

Window Length

	Aggregate Percentiles									
	[0.01,0.10)	[0.10,0.20)	[0.20,0.30)	[0.30,0.40)	[0.40,0.50)	[0.50,0.60)	[0.60,0.70)	[0.70,0.80)	[0.80,0.90)	[0.90,0.99)
w=50	0.961	0.987	0.989	0.993	0.997	0.997	0.993	0.984	0.975	0.954
w=100	1.003	0.997	0.998	0.994	0.996	0.995	0.992	0.987	0.984	0.973
w=200	1.016	1.009	1.009	1.004	1.002	1.001	0.998	0.995	0.994	1.001



Aggregate quantile loss relative to the DMQ and backtest p-values (unconditional left, conditional right) for CP1 with $w = 50, 100, 200$.

Longer windows better in the unconditional test, shorter windows in the conditional test.

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